CV

GENERAL INFORMATION

Name	Hilda Sandström
Position	Marie Skłodowska-Curie postdoctoral researcher
Institution	Technical University of Munich, Germany
Email	hilda.sandstroem@tum.de
ORCID	0000-0001-7845-1088
Google Scholar	scholar.google.com

CORE COMPETENCES

Scientific leadership · Project management · Molecular modelling & simulation · Structure prediction · Cheminformatics · Machine learning for chemistry · High-performance computing (HPC) · Student supervision & mentoring · Interdisciplinary collaboration · Scientific communication

EXPERIENCE

Since 10/2025

Marie Skłodowska-Curie postdoctoral researcher

Technical University of Munich, Germany

Main project: Machine learning–based compound identification with mass spectrometry

- Developed machine learning models for mass spectrometry signal prediction and dataset similarity analysis.
- Develop protocols for simulating mass spectrometry data for atmospheric compounds.
- Coordinated interdisciplinary projects and supervised students.

9/2024 – 9/2025

Visiting postdoctoral researcher

University of Gothenburg, Sweden

Simulated mass spectrometry signals using machine learning models, molecular dynamics, reaction exploration and quantum chemistry.

9/2022 – 9/2025

Postdoctoral researcher

Aalto University, Finland

Main project: Machine learning–based compound identification with mass spectrometry

- Developed machine learning models for mass spectrometry signal prediction and dataset similarity analysis.
- Designed molecular descriptors enabling interpretable machine learning models.
- Benchmarked models and descriptors for reaction rate prediction.
- Coordinated interdisciplinary projects and supervised students.

9/2017 – 5/2022

Early-stage researcher (PhD)

Chalmers University of Technology, Sweden

Main project: Kinetic modeling and molecular structure prediction in polymerization reactions

- Applied steered molecular dynamics, density functional theory, umbrella sampling, and metadynamics for reaction pathway exploration and free-energy profiling.
- Predicted crystal structures of molecular co-crystals and identified plausible reaction products from kinetics/thermodynamics.
- Coordinated multi-site collaborations on crystal structure prediction and lipid conformer analysis; advised students.

EDUCATION

9/2017 – 5/2022

PhD in Chemistry (Theoretical chemistry)

Chalmers University of Technology, Sweden

Award date 02/06/2022. Thesis: *Nitriles in Prebiotic Chemistry and Astrobiology*. Supervisor: Prof. Martin Rahm.

8/2012 – 9/2017

MEng in Chemical engineering with engineering physics

Chalmers University of Technology, Sweden

Award date 08/11/2017.

8/2015 – 9/2017

MSc in Engineering physics (Nanotechnology master program, integrated)

Chalmers University of Technology, Sweden

Award date 08/11/2017.

8/2012 – 6/2015

BSc in Chemical engineering with engineering physics (integrated)

Chalmers University of Technology, Sweden

Award date 12/06/2015.

TEACHING

LECTURES AND EXERCISES

2018–2020

Quantum engineering
Chalmers University of Technology
Computer labs · 1st year MSc Nanotechnology · 2 h/week

2018–2021

Physical chemistry
Chalmers University of Technology
Tutorials and experimental labs · 2nd year BSc Biotechnology · 12 h/week

2018–2021

Theoretical chemistry
Chalmers University of Technology
Computer labs · 3rd year BSc Chemical engineering · 4 h/week

2017–2018

Chemistry and biochemistry
Chalmers University of Technology
Experimental labs · 1st year BSc Chemical engineering · 8 h/week

2014

Calculus
Chalmers University of Technology
Exercise sessions · 1st year BSc Chemical engineering · 1 h/week

PEDAGOGICAL TRAINING

2019

Teaching, learning and evaluation
Chalmers University of Technology
3 ECTS

SUPERVISION OF STUDENTS

Since 2024

Supervisor of MSc student — Aalto University

Since 2024

Advisor of PhD student — Aalto University

Since 2022

Co-supervisor of PhD student — University of Helsinki

11/2024 – 5/2024

Supervisor of BSc student — Aalto University

5/2021 – 9/2021

Co-supervisor of 2 visiting and 3 BSc students — Chalmers University of Technology

1/2021 – 6/2021

Co-supervisor of 6 BSc students — Chalmers University of Technology

6/2020 – 8/2020

Supervisor of 2 BSc students — Chalmers University of Technology

1/2020 – 6/2020

Supervisor of 6 BSc students — Chalmers University of Technology

4/2019 – 7/2019

Supervisor of visiting BSc students — Chalmers University of Technology

4/2018 – 6/2018

Supervisor of one BSc student — Chalmers University of Technology

SKILLS AND COMPETENCES

PROGRAMMING

Python, MATLAB, Bash — Well experienced

MACHINE LEARNING AND CHEMINFORMATICS

Scikit-learn, TensorFlow, RDKit, OpenBabel, ASE — Experienced

MOLECULAR DYNAMICS AND SIMULATION

CP2K, GROMACS, PLUMED — Expert · xTB, QCxMS, VMD — Experienced

HIGH-PERFORMANCE COMPUTING (HPC)

Parallel computing, cluster resource management — Experienced

VERSION CONTROL

Git — Experienced

LANGUAGES

Swedish (Excellent) · English (Excellent) · Italian (Intermediate) · French (Basic)

AWARDS AND HONOURS

2025

Marie Skłodowska-Curie postdoctoral fellowship
202,000 EUR

2024–2025

LUMI extreme scale access resource allocation

2018–2021

Travel grants
Nils Philblad Foundation (2021) · Karl and Annie Leon's Foundation (2018–2019)

ACADEMIC SERVICE

2026

Thesis reviewer and opponent — Universitat Autònoma de Barcelona (UAB), Barcelona, Spain

2025

Reviewer for *ACS Earth Space Chem*, *ACS Omega* and *Atmospheric Chemistry and Physics*

2025

Organizing committee — Nordic Workshop on AI for Climate Change, Sweden

2025

Core member, organizer and Finland representative — Climate AI Nordics Network

2024

Panelist on AI in chemistry, physics, and education — FysKemDagarna

2023

Organizer of workshop hands-on session — Shaking Up Tech, Workshop for underrepresented groups in STEM, Aalto University, Finland

2023

Session chair and organizer — ESTML, Levi, Finland

2022

Session chair — AbSciCon, USA

PEER-REVIEWED PUBLICATIONS

15 peer-reviewed articles, 6 first author. Total citations: 141, h-index: 5, i10-index: 5 — Google Scholar, updated 2 April 2026

15. Cappelletti, M., **Sandström, H.**, & Rahm, M. *ACS Central Science*, 12, 111–121 (2026). DOI: 10.1021/acscentsci.5c01497.
14. Lind, L., **Sandström, H.**, & Rinke, P. *The Journal of Chemical Physics*, 164 (2026). DOI: 10.1063/5.0308548.
13. Madan, I., Aliabadi, S. A., Huhtasaari, J., Matic, E., Hogedal, E., Kamińska, K., Nilsson, F., Stark, A., Izquierdo-Ruiz, F., **Sandström, H.**, Rahm, M. *QRB Discovery*, 6, e23 (2025). DOI: 10.1017/qrd.2025.10012. *[Supervised students and co-created workflow for testing stability of polymers.]*
12. Brean, J., Bortolussi, F., Rowell, A., Beddows, D. C. S., Weinhold, K., Mettke, P., Merkel, M., Kumar, A., Barua, S., Iyer, S., Karppinen, A., **Sandström, H.**, Rinke, P., et al. *ACS ES&T Air*, 2, 1704–1713 (2025). DOI: 10.1021/acsestair.5c00119. *[Supervised PhD student F. Bortolussi in developing and evaluating the machine learning model and workflow.]*
11. Izquierdo-Ruiz, F., Cable, M. L., Hodyss, R., Vu, T. H., **Sandström, H.**, Lobato, A., & Rahm, M. *Proc. Natl. Acad. Sci. U.S.A.*, 122, e2507522122 (2025). DOI: 10.1073/pnas.2507522122. *[Developed and tested crystal structure prediction program workflow for molecular cocrystals.]*
10. Valiev, R. R., Nasibullin, R. T., **Sandström, H.**, Rinke, P., Puolamäki, K., & Kurten, T. *Physical Chemistry Chemical Physics*, 27, 14804–14814 (2025). DOI: 10.1039/D5CP01101A. *[Co-advisor for ML workflow; developed MBTR model.]*
9. Bortolussi, F., **Sandström, H.**, Partovi, F., Mikkilä, J., Rinke, P., & Rissanen, M. *Atmospheric Chemistry and Physics*, 25, 685–704 (2025). DOI: 10.5194/acp-25-685-2025. *[Co-designed study, advised, and contributed to programming and model testing.]*
8. Malaska, M. J., **Sandström, H.**, Hofmann, A. E., Hodyss, R., Rensmo, L., van der Meulen, M., Rahm, M., Cable, M. L., & Lunine, J. I. *Astrobiology*, 25, 367–389 (2025). DOI: 10.1089/ast.2024.0125. *[Performed geometry optimizations, conformer search and student supervision.]*
7. **Sandström, H.**, & Rinke, P. *Geoscientific Model Development*, 18, 2701–2724 (2025). DOI: 10.5194/gmd-18-2701-2025.
6. **Sandström, H.**, Rissanen, M., Rousu, J., & Rinke, P. *Advanced Science*, 11, 2306235 (2024). DOI: 10.1002/advs.202306235.
5. **Sandström, H.**, Izquierdo-Ruiz, F., Cappelletti, M., Dogan, R., Sharma, S., Bailey, C., & Rahm, M. *ACS Earth and Space Chemistry*, 8, 1272–1280 (2024). DOI: 10.1021/acsearthspacechem.4c00088.
4. **Sandström, H.**, & Rahm, M. *The Journal of Physical Chemistry A*, 127, 4503–4510 (2023). DOI: 10.1021/acs.jpca.3c01504.
3. **Sandström, H.**, & Rahm, M. *ACS Earth and Space Chemistry*, 5, 2152–2159 (2021). DOI: 10.1021/acsearthspacechem.1c00195.
2. **Sandström, H.**, & Rahm, M. *Science Advances*, 6, eaax0272 (2020). DOI: 10.1126/sciadv.aax0272.
1. Lindblom, A., Sriram, K. K., Müller, V., Öz, R., **Sandström, H.**, Åhrén, C., Westerlund, F., & Karami, N. *Diagnostic Microbiology and Infectious Disease*, 93, 380–385 (2019). DOI: 10.1016/j.diagmicrobio.2018.10.014. *[Performed fluorescence microscopy assays where I stained, trapped, and photographed plasmids in nanochannels.]*

TALKS

INVITED SEMINARS AND KEYNOTES

2026

Data-driven compound identification with atmospheric mass spectrometry
Network on Mathematical Data Science for Materials Science, Workshop on the Interface of Mathematics and Machine Learning for Applications in Materials Science — University of Glasgow, UK

2025

CLOUDMAP – Advanced identification of atmospheric compounds
*Atmospheric day, Sweden — **Keynote***

2025

Machine learning for atmospheric mass spectrometry
Nordic Workshop on AI for Climate Change, Sweden

2024

AI in Chemistry: Solving experimental challenges with artificial intelligence
FysKemDagarna (Physics and Chemistry Days), Sweden

CONTRIBUTED TALKS

2026

Towards atmospheric compound identification using simulated electron ionization mass spectra
Chemical Compounds Space Conference (CCSC 2026), Munich, Germany

2023

Characterizing Atmospheric Molecules for Machine Learning
International Aerosol Modeling Algorithms Conference, USA

2023

Characterizing Atmospheric Molecules for Machine Learning
European Aerosol Conference, Spain

2023

Characterizing atmospheric molecules for machine learning
Physics Days, Finland

2022

Untangling hydrogen cyanide polymerization using quantum chemistry
AbSciCon, USA